

Typical Clustering Methodologies

- ❑ Distance-based methods
 - ❑ Partitioning algorithms: K-Means, K-Medians, K-Medoids
 - ❑ Hierarchical algorithms: Agglomerative vs. divisive methods
- ❑ Density-based and grid-based methods
 - ❑ Density-based: Data space is explored at a high-level of granularity and then post-processing to put together dense regions into an arbitrary shape
 - ❑ Grid-based: Individual regions of the data space are formed into a grid-like structure
- ❑ Probabilistic and generative models: Modeling data from a generative process
 - ❑ Assume a specific form of the generative model (e.g., mixture of Gaussians)
 - ❑ Model parameters are estimated with the Expectation-Maximization (EM) algorithm (using the available dataset, for a maximum likelihood fit)
 - ❑ Then estimate the generative probability of the underlying data points
- ❑ High-dimensional clustering

High-dimensional Clustering

- ❑ Subspace clustering: Find clusters on various subspaces
 - ❑ Bottom-up, top-down, correlation-based methods vs. δ -cluster methods
- ❑ Dimensionality reduction: A vertical form (i.e., columns) of clustering
 - ❑ Columns are clustered; may cluster rows and columns together (co-clustering)
 - ❑ Probabilistic latent semantic indexing (PLSI) then LDA: Topic modeling of text data
 - ❑ A cluster (i.e., topic) is associated with a set of words (i.e., dimensions) and a set of documents (i.e., rows) simultaneously
 - ❑ Nonnegative matrix factorization (NMF) (as one kind of co-clustering)
 - ❑ A nonnegative matrix A (e.g., word frequencies in documents) can be approximately factorized two non-negative low rank matrices U and V
 - ❑ Spectral clustering: Use the spectrum of the similarity matrix of the data to perform dimensionality reduction for clustering in fewer dimensions